

ROHAN AJAY VARGHESE

Software Engineer, M.Sc. Industrial Artificial Intelligence

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EDUCATION

Hochschule Albstadt-Sigmaringen
M.Sc. Industrial Artificial Intelligence
March 2026 - present

CHRIST University, Bangalore, India
B.C.A. Computer Applications (3.5/4.0)
2022 - 2025

Outskill
AI Accelerator Programme
2026

SKILLS

- **Languages**: Python, JavaScript, TypeScript, Java
- **AI and ML:** RAG, Large Language Model (LLM) Integration, LlamaIndex, LangGraph, Qdrant, Embeddings, Hybrid Search, LLM Evaluation, OCR, Prompt Engineering, Machine Learning, n8n
- **Frontend**: React.js, PrimeReact, WebSockets, React Native, Expo, Tailwind CSS, Vite
- **Backend and Data**: FastAPI, REST APIs, Firebase Firestore, MySQL, Pandas, NumPy
- **Cloud and Tools**: Google Cloud Run, Docker, AWS, Git, GitHub, SonarQube, Figma

LANGUAGE

English (fluent)

German (A2)

ABOUT ME

Master's student in Industrial AI (current average 1.4). In India, I spent a year as a Frontend Engineer at Viamagus Technologies, building screens and turning proofs of concept into shipped features for client products in oil and gas, scientific research and warehouse automation. In Germany since March 2026, I completed a semester project with a medical technology company evaluating autonomous mobile robot (AMR) vendors, and built MyUniGuide, a retrieval-augmented generation (RAG) system, and Claude Vitals, an open-source desktop tool.

EXPERIENCE

Frontend Engineer

Viamagus Technologies, Bengaluru, India 2025 - 2026

Oil and Gas Virtual Reality (VR) Platform *React, PrimeReact, Pano2VR*

- Built new screens and reusable components in React and PrimeReact, including home and dashboard screens, data-table filters, report lists and draggable detail panels.
- Delivered 3D model tools (ViewCube, selector, properties) and simultaneous operations (SIMOPS) permit highlights with drill-down.
- Built proofs of concept, such as live worker tracking with zone-based safety alerts and smooth asset zoom, that later became product features.
- Built the UI for a voice and text AI assistant and a signal-mapping module with Excel import; resolved about 255 SonarQube warnings.

Remote Scientific Experiment Platform *React, Blockly*

- Built the core screens of a Blockly-based experiment platform: dashboards, experiment creation, project management, profile and settings.
- Added the Blockly toolbox, icon sidebar, validated sign-in flows, confirmation modals, loading skeletons and toast notifications.

AMR Warehouse Automation System *React, WebSockets*

- Built new operator screens for real-time mission tracking over WebSockets, with timelines, status monitoring and robot navigation controls.
- Fixed rendering bugs in the robot fleet visualizer.

PERSONAL PROJECTS

MyUniGuide (V1 and V2)

Bilingual RAG system 2026

V1: RAG pipeline built from scratch over my programme's four regulations, including German and scanned PDFs. Matched a long-context baseline at 84% on 50 questions while catching contradictions it missed. V2: Rebuilt with LlamaIndex, Qdrant and Gemini over 19 regulations, with hybrid retrieval, German and English answers and authority-aware conflict detection. 90.3% on 72 questions, zero fabricated answers, 23 times cheaper than the baseline. Deployed on Google Cloud Run.

Claude Vitals

Open-source desktop tool 2026

Python Windows tray app that monitors Claude usage limits with colour-coded status and alerts. Local-first, tested and released open source under the MIT License.

RoomEase

Roommate Finding App 2024

Roommate matching app built with React Native and Firebase.

InsightFlow

Inventory Management System 2024

Inventory and sales dashboards in Python, Pandas and Plotly.